Drone Initiatives of Various Organizations

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ABSTRACT

Commonly used in military settings, unmanned aerial vehicles have made their appearance in civilian applications across a variety of different categories. In response to the trending potential of drones for commercial purposes the Federal Aviation Administration has recently approved drones for movie making.[1] This motion has opened up questions about the future of drones and its inevitable integration into civilian airspace.

Many organizations are exploring use cases of drones in areas such as mission critical search and rescue, quick response delivery for medicinal or health related purposes, and much more when it comes to emergency applications. Other initiatives include universal Internet access, spreading of pesticides in agricultural settings, geographical mapping and cargo delivery.

This paper will serve to exemplify and shed light on unmanned aerial vehicles in the civilian world. A broad perspective will be taken to understand the current initiatives and also the initiatives moving forward.

Drones have been subject to stereotypical classification as a dangerous, murderous tool that governments would use during times of war or as surveillance for intervention. The idea of having this type of system deployed as a benefactor will widen their intended use and change the way we shop, work, send and receive mail or packages and even introduce a new way to capture media.

In addition to illustrating organizational efforts to bring drones into the scope of civilizations, standards need to be discussed and explained to provide a foundation for future references and integration. Defining a common ground for all types of drones will help during emergency drone conditions which can include bad weather, drone-to-drone communication for collision avoidance and other operational hazards.

Keywords

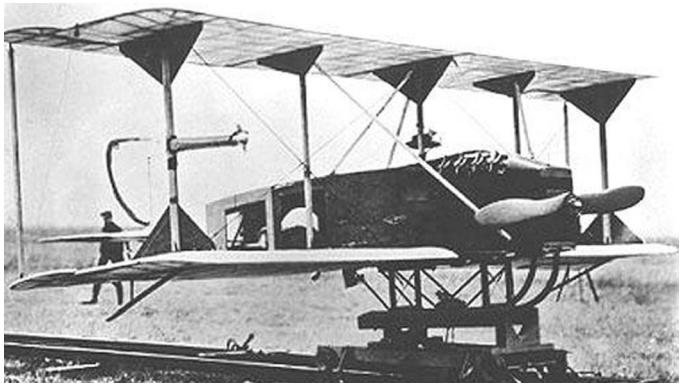
Unmanned aircraft systems, unmanned aerial vehicles, drones, civilian applications, initiative of organizations, standards, foundation

1. NOT-SO-POPULAR DRONES

1.1 A bit of history

The earliest form of an unmanned aerial vehicle dates back to August 22nd, 1849 when Austrians, who controlled most of Italy at the time, launched an unmanned balloon raid assault on Venice equipped with explosives that was copper wire fed to drop the bombs during favorable wind conditions.[2] Although the modern definition of a UAV does not necessarily include balloons, the concept became a foundation for current combat drones.

The next iteration of pilot-less aircraft, with wings and radio control, was in World War I when they were being developed to target zeppelins which were dominantly used for bombing and scouting as well as commercial transportation later. On September 12th, 1916, the Hewitt-Sperry Automatic Airplane made its appearance as the “flying bomb” and demonstrated the initial idea behind an unmanned aircraft.[3]



Photograph 1: The Hewitt-Sperry Automatic Airplane with radio control developed during World War I.[4]

It is important to note that in 1898 Nikola Tesla made remote control possible via radio waves and that inventors around the world tinkered with Tesla's new innovative idea; particularly Elmer Sperry, a naval gyroscope maker who had conceived the creation of the Automatic Airplane.[5] It wasn't until the 1930s and early to mid 1940s that the notion of a military unmanned aircraft used for war became subconsciously linked to its civilian counterparts.

1.2 Drone sentiment

Around 1934, Reginald Denny (an actor and radio control model aircraft enthusiast) formed "Reginald Denny Industries" and opened a model plane shop on Hollywood Boulevard known as "Reginald Denny Hobby Shops".[6] The shop evolved into the "Radioplane Company" where Denny believed that low-cost RC aircraft can be used for training anti-aircraft gunners and demonstrated a prototype target drone in 1935. This became the first large-scale production of purposefully built drones.

He then bought a design from Walter Righter in 1938 and began marketing the "Dennymite" to hobbyists and demonstrated military applications of the "Dennymite" to the United States Army. Fifteen thousand drones were made for the Army and it was at the Van Nuys Radioplane factory that in 1944 a photographer, David Conover, saw a young lady named Norma Jeane who he said had the potential to be a model. This led to fame for Jeane, who changed her name to Marilyn Monroe and made many headlines including her *Yank* magazine headline and photograph in 1945.

Photograph 2: Marilyn Monroe, a technician at the Radioplane munitions factory, photographed at her job in 1945.[7]

With her wide spread success, Marilyn Monroe became a World War II female worker poster model usually featuring her close ties to aviation by taking photographs of her travels as she exited or entered aircraft. This is the first time that the building blocks of the unmanned aerial vehicles we know of today received mass publicity. Further inventions for radio controlled aircraft helped propel their intended use to different settings and also into more practical civilian uses.

One such invention was radio controlled aircraft that could fly out of sight. This lead to the invention of the aerial torpedo which later included RCA television cameras and a television screen in a nearby control aircraft. Although many military applications of UAVs were effective, it became a conflict and an expense during times of war. Disagreements arose on using drones that can easily be salvaged by an enemy country and reverse engineered. In addition, government expenditures limited its scope when it came to combat drones, and it was determined that other means may be more effective.

By the 1950s, piloted combat aircraft were capable of Mach 2 and more development went into making faster target drones. B-17 aircraft were also transformed into drones for collecting radioactive data where they were controlled from a jeep during takeoff and landing and also from a transmitter on a separate B-17 during flight. Instead of combat intense uses, drones now steered more toward passive applications.

At around the same time and up until 2006, US drones were used for reconnaissance which were applied to spy on North Vietnam, Communist China, and North Korea. The Vietnam War became a major factor for further drone use and was deployed regularly for reconnaissance as well as other strategic counter measures. Battlefield UAVs took rise in the 1980s with the Israeli Air Force's victory over the Syrian Air Force. Israel used the drones as electronic decoys, electronic jammers and for real time video reconnaissance.[8]

In the 80s, new attention focused on aircraft propulsion and sustainability via solar power. Many "solar challenges" arose and ultimately lead to advancements in aviation but also concluded that the technology was not up to par with this feat. NASA awarded a contract to Lockheed Martin to study a solar-powered UAV for missions such as crop monitoring, military reconnaissance, and communications relay. This concept was also ahead of its time and efforts were focused on more conventional concepts.

Later, DARPA issued a $40 million US contract to Leading Systems Incorporated to build an endurance UAV named "Amber" which was to be used as a multipurpose military UAV. Amber was successful but funding for reconnaissance assets were being cut and eventually lead to the bankruptcy of the company. In 1991, General Atomics bought out Leading Systems Incorporated who later developed Amber into an operational platform known as the MQ-1 Predator.[9]

Although successful as a military UAV, the Predator was also the first UAV to be integrated into civilian airspace. Its multiple uses served in police force work which was used to survey the US and Mexican border for illegal immigrants and drugs.[10]

1.3 Practical drones

In 2005, requests had been made for the use of the Predator aircraft to be used in search and rescue operations following hurricane Katrina, but the Federal Aviation Administration did not have authorization in place and the assets were not used. The Predator's infrared camera, that can zoom with digital enhancement, has the capability of isolating heat signatures of a human body from an altitude of 10,000 feet, making it an ideal search and rescue tool.

Finally, on May 18th, 2006, the FAA issued a certificate of authorization to allow different variants of the Predator UAV to be used within the United States civilian airspace for the search of disaster survivors. In a *Wall Street Journal* report it was stated that, "After distinguished service in war zones in recent years, unmanned planes are hitting turbulence as they battle to join airliners and weekend pilots in America's civilian skies. Drones face regulatory, safety and technological hurdles – even though demand for them is burgeoning. Government agencies want them for disaster relief, border surveillance and wildfire fighting, while private companies hope to one day use drones for a wide variety of tasks, such as inspecting pipelines and spraying pesticides on farms."[11]

2. RISE OF THE DRONES

2.1 Sparks of a trend

With drones becoming more transparent, countries and companies alike are being pressured by its growing popularity. Both military and civilian headlines contributed to a sharp trend illustrated by the Google Trends map below:

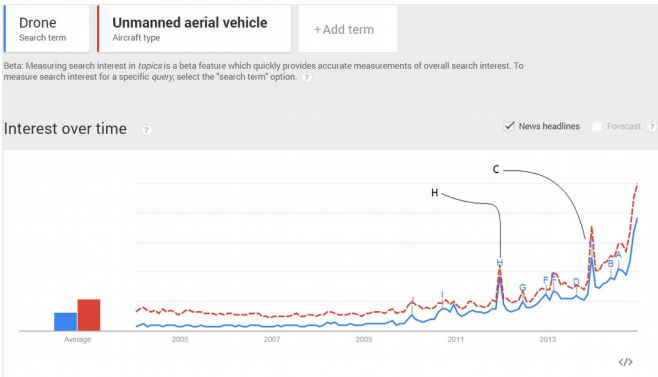
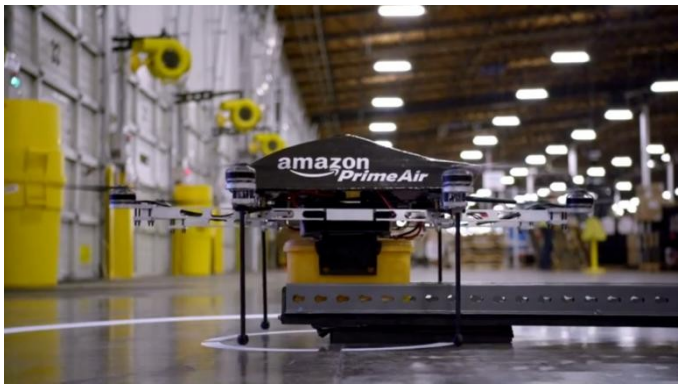


Figure 1: A trend chart showing the relation of terms 'drone' and 'unmanned aerial vehicle'.[12]

The first spike, labeled H, was when the United States lost a drone to Iran and was asking for it back. Iran was able to successfully hack into the drone and control it. The drone landed on Iranian territory where they were able to reverse engineer the drone and decode the data off of it. The proceeding trend headlines included separate drone attacks causing both military and civilian casualties. The second spike, labeled C, was what really brought attention to drones and caused a further rally.

2.2 Amazon starts the fire

On December 1st, 2013, Amazon unveiled its plan to revolutionize delivery. Approximately 86% of their products weigh under 5 pounds and can get from the warehouse to a persons doorstep in under 30 minutes using Amazon Prime Air Drones.[13] The motors on the quadrocoptor are powered entirely off of an electric battery on board the drone. This will be a more fuel efficient delivery mechanism for delivery travel up to 10 miles. The system is also meant to be fully automated.



Photograph 3: A quadrocopter drone known as the Amazon Prime Air drone. Attached is a small yellow package getting ready for shipment.[14]

A clip from the television show known as *60 Minutes* called *Drones over America* depicted its continued growth by casting the show on the floor of an unmanned aerial vehicle convention. Companies such as Boeing, Lockheed Martin and other military organizations set up booths to show off their version of a drone. A helicopter was completely transformed into a remote control search and rescue drone that would be able to pick up injured soldiers without endangering a pilots life.[15] Another drone assembled from pieces of Styrofoam was able to scout out potential enemy targets and weighed about 2 pounds.[16]

A more civilian application became apparent when transitioning the simple use of a drone for photography to using a drone for photographing plants in a farmland and determining which areas needed more water and care. Some drones were shown to be used for wild fires and measuring catastrophes. Drones assisted humans in detecting radiation after the 2011 tsunami in Japan hit a

nuclear power-plant that caused a meltdown.[17] Spoofs of Netflix (delivering movies), and Dominos (delivering pizza pies) used drones to bring the future of delivery a little closer to home. More companies began to join the efforts of drone use and advocated their view on them as well.

2.3 The Google initiative

"If there is one thing Google likes, it is changing the world." *The Atlantic* article titled *Inside Google's Secret Drone-Delivery Program* captures the essence behind using drones for every day activities.[18] A social shift becomes inevitable as it is portrayed that, "It would help move us from an ownership society to an access society." "The idea goes like this: Because people can't assume near-instantaneous delivery of whatever they need, they stockpile things. They might have a bunch of batteries, slowly decharging in a drawer, or a drill that they use for 10 minutes a year. Each of these things is a personal possession that sits around, embodying all this energy and industrial effort unproductively." We begin to share things more and are drawn to a changing environment that has yet to be experienced. This community feeling, lowering of noise pollution and our carbon footprint while improving safety of this endeavor is precisely the idea behind Google's Project Wing.

One of a few Google X projects, Wing is a competing program with Amazon Prime Air with Google's own perspective. The drone, a tail-sitting aircraft, takes a package attached to it, flies to the destination and uses a winch to drop the package mid flight and slow it before it hits the ground. It releases the winch and travels back to its origin for a new round of packages. "There is an opportunity to solve a big problem and I think Google's announcement will[sic] accelerate that even further." said Raptopoulos of Matternet.

Photograph 4: Google's Project Wing drone. Attached is a package with a winch to the drone for dropping and releasing its payload.[19]

Another Google X project known as Project Loon aims to bring Internet access to everyone by interconnecting balloons in a network flying above weather and all aircraft. This approach may be the most viable since nothing (except for space bound rockets or ships) will be disrupting the balloon-net. It will effectively give access to a communication network for everyone to use and to those who are in need of help.

Photograph 5: Google's Project Loon balloon that is meant to reside in the stratosphere which will connect everyone to the Internet.[20]

2.4 Facebook makes a move

Similar to Project Loon, Facebook is developing a network of solar-powered drones in conjunction with satellites to provide Internet access to the world. According to the BBC News article *Facebook Drones to offer low-cost net access,*[21] the social network hired a five-member team that worked at British firm Ascenta, whose founders developed a drone that holds the record for the longest-flying solar-powered unmanned aircraft. Facebook's project seems much more feasible with current technology since it would be adding to already existing sources.

Unfortunately for Facebook the matter became more difficult when Google decided to buy out the drone company Facebook wanted to buy for its fleet.[22] Titan Aerospace as well as the Makani airborne wind turbine power company that Google acquired a year before, will be working together on Project Loon. However, Facebook bought out Ascenta and isn't out of the game just yet. In fact, Mark Zuckerberg vowed that Facebook will shoot down Google drones.[23] He announced plans to build a $24 billion Facebook laser shield, a global network of satellites capable of identifying and incinerating Google drones in midair.

Within an hour of Zuckerberg's announcement, Google responded with a stern warning of its own, vowing, "Any act of aggression against Google will

not stand." Google announced that it was prepared to shoot down Facebook's laser satellites with a long-range super cannon called Google Gun.

Although the competition between the two companies was more of a comic, both are involved in an effort known as internet.org. Facebook established this to bring the technology community together in a joint program to provide cheaper Internet access to everyone. Even though the coalition helps to solve a world problem, to connect everyone, the companies involved will need to come up with all sorts of technology to bridge the gap.

Figure 2: Facebook's project to connect the world to the internet.[24]

2.5 DHL chimes in

Although Amazon and Google have been testing their new drone delivery systems, German company Deutsche Post DHL has already been carrying out drone missions to collect data about how unmanned aerial vehicles behave in special weather conditions.[25] "Its project represents the first and only time to date that drones are carrying out automated real-life missions in Europe." The 'parcelcopter' flew from one end of the Rhine River to the other approximately 50 times. After the analysis, the company wanted the drone to endure longer missions and heavier weather conditions. In its second round, DHL had the drone deliver medical supplies from a city to a remote island. With harsher weather conditions, DHL was able to collect compelling data to push the project further. Although it was stated that DHL does not plan on helping the tech giants; "We kept the information [of their mission's findings] to ourselves because we don't want to share with Amazon and Google."[26]

2.6 The push for open source

Skycatch, DroneDeploy, HobbyKing, Horizon Ag, PrecisionHawk, Agribotics and Walkera all have one thing in common. They are among the drone designers already using Dronecode.[27] The Linux Foundation has developed Dronecode to be the new open source standard of drone software to encourage cooperation of the peaceful use of drones.[28] It brings together existing open source code for unmanned aerial vehicles under a non-profit governance system. The numbers are staggering, with 1,200 developers working on the newly aligned projects and over 150 code commits per day. "Unfortunately when most people think of drones they think of military use, but drones are being used in a variety of cool, exciting ways – agriculture, search and rescue, realtime mapping, construction," "Creating the software stack by yourself seems a little bit crazy! The Linux Foundation is a place where we can grow these type of software communities." The newly created website and further information can be found at:

https://www.dronecode.org/

3. CONSEQUENCES

While ambition grows following the trend of drones in civilian skies, it is important to explain the downfalls of using them as well. Varying weather conditions deter drone use and birds can easily take them down without a moments notice. An article titled *Beware! 8 sinister consequences of Google and Amazon drones*, illustrate some comical but also real scenarios.[29]

"You will never be alone." Already a hot issue, if we were to become accustomed to drones flying around, they would suddenly be much more useful for inconspicuous surveillance.

"They will be hacked." With consistent data feeds of drone communication, someone, somewhere is going to figure out how to hack into them and either get access to personal information or take over the drone.

"Drone hunting." One town is already considering issuing a license to hunt drones. This may be a full-blown hobby for reactionaries and

technophobes with only a marginal amount of good judgment and common sense.

4. CONSIDERATIONS

Not entirely complete, the following points are personal considerations for the future of drones and their applications. These points may already exist and should be stated for reference in case anyone else would like to expand upon this research.

-Federal website devoted to a network that is instantiated via drone routing communication with said website.

-Common hardware such as the traffic collision and avoidance system currently mandated on small to commercial aircraft.

-A drone framework or handbook that can be used as a reference.

-Follow up bulletin of the unmanned aircraft systems road-map.[30]

-A standard for drone communication used to interchange data and automated air traffic control via a medium such as cell towers.

-Accepted or considered categories for evaluation and propagation of new uses for drones that would speed up their existence in the space.

-A committee devoted to determining different facets of drone use and conception.

5. CONCLUSION

The inevitable integration of drones for civilian use continues to trend and will eventually be incorporated. Starting from understanding the history behind unmanned aerial vehicles to efforts made by companies in exploiting benefactors, drones will need a place for the future to come. The Federal Aviation Administration needs to make haste in securing funding as well as devoting time and energy in structuring this newly disruptive sector. With open source in mind, much can be done with the already enthusiastic crowd of people looking to set this in motion. All major companies are vested in supporting the change and also continuing the incline of drone use.

Get ready for 'Drone Nation', is an article published by Fortune to really pin down its current impact on the world.[31] It is stated that the global market for nonmilitary drones has already ballooned into a $2.5 billion industry and it's growing 15% to 20% annually. This is all under the current law. Japan, Australia, France, and in the UK, UAVs for commercial purposes are soaring. A small company, AirDroids, has already received $1.2 million in orders for its compact UAV and this is just the tip of the iceberg. “There's this gray area with rules, but people are going ahead and using these things.”

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